

5. Laplace Transform

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5.1 Introduction

The method of solving a differential equation involves many steps, namely

(a). Computation of complementary function
(b). Computation of particular integral
(c). Evaluation of the arbitrary constants using the initial conditions.

But Laplace transform is a powerful tool for solving differential equations without involving all the above steps. By the application of Laplace transforms, the given differential equation can be converted into an algebraic equation and the process of finding the solution becomes easy.

Laplace transforms was first used by the French Mathematician Pierre Simon De Laplace in the 18th century. Laplace transform has wide applications in the Engineering fields like network analysis, control systems, circuit theory and many other areas.

Definition 5.1.1 — Laplace Transform. Let $f(t)$ be defined for all $t \geq 0$, then $\int_0^\infty e^{-st} f(t)dt$ is defined as the Laplace transform of $f(t)$, if the integral exists.

It is denoted by $L[f(t)]$. It is a function of s. Hence,

$$L[f(t)] = \int_0^\infty e^{-st} f(t)dt = F(s).$$

Result. The parameter s may be real or complex.

5.2 Sufficient condition for existence of Laplace Transforms

Let f(t) be defined for all $t > 0$ such that (i) f(t) is piecewise continuous in the interval $[0, \infty)$ and (ii) f(t) is of exponential order $\alpha > 0$, then the Laplace transform of f(t) exists for $s > \alpha$.

Note 5.2.1 1. Picewise continuous on $[0,\infty)$ means that the function is continuous on every finite sub interval $0 \le t \le \alpha$ except possibly at a finite number of points where they have jumps i.e., f(x+) and f(x-) exist but not equal.

2. $f(t)$ is of exponential order $\alpha > 0$ if $|f(t)| \le Me^{\alpha}t$ for all $t \le 0$ and M is a constant. In other words, $\lim_{t\to\infty}(e^{-\alpha t} f(t))$.

Example : t^n is an exponential order as $t \to \infty$

$$\lim_{t\to\infty}\left(e^{-\alpha t}t^n\right) = \lim_{t\to\infty}\left(\frac{t^n}{e\alpha t}\right) = \lim_{t\to\infty}\left(\frac{n!}{\alpha^n e^{\alpha t}}\right)$$

Result The above conditions are sufficient but not necessary.

Problem $L\left[\frac{1}{\sqrt{t}}\right]$ exists but it is not continuous at $t = 0$.

Definition 5.2.1 — Linearity Property. If $f(t)$ and $g(t)$ are two continous functions of t and k is a constant, then

(a). $L[f(t) \pm g(t)] = L[f(t)] \pm L[g(t)]$

(b). $L[kf(t)] = kL[f(t)] where k \neq 0.$

5.2.1 Laplace Transforms of standard functions

1. $L[1] = \int_0^\infty e^{-st}1dt = \left[\frac{e^{-st}}{-s}\right]_0^\infty = -\frac{1}{s}[0-1] = \frac{1}{s}$
2. $L[e^{at}] = \int_0^\infty e^{-st}e^{at}dt = \int_0^\infty e^{-(s-a)t}dt = \left[\frac{e^{-(s-a)t}}{-(s-a)}\right]_0^\infty = -\frac{1}{s-a}[0-1] = \frac{1}{s-a}$
3. $L[e^{at}] = \frac{1}{s+a}$
4. $L[t] = \frac{1}{s^2}$
5. $L[t^n] = \frac{n!}{s^n+1}$
6. $L[\cosh at] = \frac{s}{s^2-a^2}$
7. $L[\sin hat] = \frac{a}{s^2-a^2}$
8. $L[\cos at] = \frac{s}{s^2+a^2}$
9. $L[\sin at] = \frac{a}{s^2=a^2}$

5.2.2 WORKED PROBLEMS

Problem 5.1 Find $L[\sin 2t \sin 3t]$.

Solution. $L[\sin 2t \sin 3t] = \frac{1}{2}L[\cos t - \cos 5t] = \frac{1}{2}\left[\frac{s}{s^2+1} - \frac{s}{s^2+25}\right] = \frac{12s}{(s^2+1)(s^2+25)}.$ ■

Problem 5.2 Find $L[\sin^3 2t]$

Solution. $L[\sin^3 2t] = L\left[\frac{1}{4}(3\sin 2t - \sin 6t)\right] = \frac{1}{4}\left[3\frac{2}{s^2+4} - \frac{6}{s^2+36}\right] = \frac{48}{(s^2+48)(s^2+36)}$ ■

Problem 5.3 Find $L[t^{-\frac{1}{2}}]$

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Solution.

$$L\left[t^{-\frac{1}{2}}\right] = \int_0^\infty e^{-st} t^{-\frac{1}{2}} dt$$

Let $st = x \to t = \frac{x}{s} \to dt = \frac{1}{s}dx$

When $t = 0, x = 0$ and when $t = \infty, x = \infty$

$$\therefore \quad L[t^{-\frac{1}{2}}] = \int_0^\infty e^{-x} \left(\frac{x}{s}\right)^{\frac{1}{2}} \frac{dt}{s} = \frac{1}{\sqrt{s}} \int_0^\infty e^{-x} x^{-\frac{1}{2}} dx = \frac{1}{\sqrt{s}} \int_0^\infty e^{-x} x^{\frac{1}{2}-1} dx$$

$$= \frac{1}{\sqrt{s}} \Gamma\left(\frac{1}{2}\right) \quad \left[\because \Gamma n = \int_0^\infty e^{-x} x^{n-1} dx\right] = \frac{1}{\sqrt{s}} \sqrt{\pi} = \sqrt{\frac{\pi}{s}}$$

■

Problem 5.4 Find $L\left[\sin^2 t \cos^3 t\right]$.

Solution.

We have $\sin^2 t \cos^3 t = \left(\frac{1-\cos 2t}{2}\right)\left(\frac{\cos 3t + 3\cos t}{4}\right)$

$$= \frac{1}{8}\left[\cos 3t + 3\cos t - \cos 2t \cos 3t - 3\cos 2t \cos t\right]$$

$$= \frac{1}{8}\left[\cos 3t + 3\cos t - \frac{1}{2}(\cos 5t + \cos t) - \frac{3}{2}(\cos 3t + \cos t)\right]$$

$$= \frac{1}{8}\left[\cos 3t + 3\cos t - \frac{1}{2}\cos 5t - \frac{1}{2}\cos t - \frac{3}{2}\cos 3t - \frac{3}{2}\cos t\right]$$

$$= \frac{1}{8}\left[-\frac{1}{2}\cos 3t + \cos t - \frac{1}{2}\cos 5t\right]$$

$$L\left[\sin^2 t \cos^3 t\right] = L\left[\frac{1}{8}\left[-\frac{1}{2}\cos 3t + \cos t - \frac{1}{2}\cos 5t\right]\right]$$

$$= \frac{1}{16} L[2\cos t - \cos 3t - \cos 5t]$$

$$= \frac{1}{16}\left[\frac{2s}{s^2+1} - \frac{s}{s^2+9} - \frac{s}{s^2+25}\right].$$

■

Problem 5.5 Find the Laplace transform of $f(t)$ defined by $f(t) = \begin{cases} e^t & 0 < t < 1 \\ 0 & t > 1 \end{cases}$

Solution.

$$L[f(t)] = \int_0^\infty e^{-st} f(t) dt = \int_0^1 e^{-st} f(t) dt + \int_1^\infty e^{-st} f(t) dt = \int_0^1 e^{-st} e^t dt$$

$$= \int_0^1 e^{-(s-1)t} dt = \left[\frac{e^{-(s-1)t}}{-(s-1)}\right]_0^1 = -\frac{1}{s-1}\left[e^{-(s-1)} - 1\right] = \frac{1 - e^{-(s-1)}}{s-1}.$$

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Theorem 5.2.2 — Shifting Property - First Shifting Theorem. If $L[f(t)] = F(s)$ then $L[e^{-at} f(t)] = F(s+a)$.

Theorem 5.2.3 — Change of scale property. If $L[f(t)] = F(s)$ then $L[f(at)] = \frac{1}{a} F\left(\frac{s}{a}\right), a > 0$

Worked Examples

Problem 5.6 Find $L[e^t t^{-\frac{1}{2}}]$

Solution. $L[e^t t^{-\frac{1}{2}}] = L\left[t^{-\frac{1}{2}}\right]_{s \to s-1} = \left[\sqrt{\frac{\pi}{s}}\right]_{s \to s-1} = \sqrt{\frac{\pi}{s-1}}$ ■

Problem 5.7 Find $L[e^{-3t} \sin t . \cos t]$

Solution.

$$L[e^{-3t} \sin t . \cos t] = L\left[e^{-3t} \frac{\sin 2t}{2}\right] = \frac{1}{2} L[e^{-3t} \sin 2t] = \frac{1}{2} L[\sin 2t]_{s \to s+3} = \frac{1}{2}\left(\frac{2}{s^2+4}\right)_{s \to s+3}$$
$$= \frac{1}{(s+3)^2+4} = \frac{1}{s^2+6s+13}$$

■

Problem 5.8 Find the Laplace Transforms of

(a). $\cosh at \sinh bt$

(b). $(1+te^{-t})^3$

Solution. (a). $L[\cosh at \sinh bt] = L[\frac{e^{at}+e^{-at}}{2} \sinh bt] = \frac{1}{2} L[e^{at} \sinh bt + e^{-at} \sinh bt]$

$$= \frac{1}{2}[L[\sinh bt]_{s \to s-a} + L[\sinh bt]_{s \to s+a}]$$
$$= \frac{1}{2}\left[[\frac{b}{s^2+b^2}]_{s \to s-a} + [\frac{b}{s^2+b^2}]_{s \to s+a}\right]$$
$$= \frac{1}{2}\left[\frac{b}{(s-a)^2+b^2} + \frac{b}{(s+a)^2+b^2}\right]$$

(b). $L[(1+te^{-t})^3] = L[1+3te^{-t}+3t^2e^{-2t}+3t^3e^{-3t}]$

$$= L[1] + 3L[te^{-t}] + 3L[t^2e^{-2t}] + 3L[t^3e^{-3t}]$$
$$= \frac{1}{s} + 3\left[\frac{1}{s^2}\right]_{s \to s+1} + 3\left[\frac{2}{s^3}\right]_{s \to s+2} + 3\left[\frac{6}{s^4}\right]_{s \to s+3}$$
$$= \frac{1}{s} + \frac{3}{(s+1)^2} + \frac{6}{(s+2)^3} + \frac{6}{(s+3)^4}$$

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Theorem 5.2.4 If $L[f(t)] = F(s)$, then $L[t^n f(t)] = (-1)^n \frac{d^n}{ds^n}[F(s)]$

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Worked example

Problem 5.9 Find $L[T\cosh 3t]$.

Solution.

$$L[t\cosh 3t] = -\frac{d}{ds}\{L[\cosh 3t]\} = -\frac{d}{ds}\left(\frac{s}{s^2-9}\right) = -\left(\frac{s^2+9-s\times 2s}{(s^2-9)^2}\right)$$
$$= -\frac{9-s^2}{(s^2-9)^2} = \frac{s^2-9}{(s^2-9)^2} = \frac{1}{(s^2-9)}$$

■

Problem 5.10 Find $L[t^2\sin at]$

Solution.

$$L[t^2\sin at] = \left(-1^2\right)\frac{d^2}{ds^2}\left(L(\sin at)\right) = \frac{d^2}{ds^2}\left(\frac{a}{s^2-a^2}\right)$$
$$= a\frac{d^2}{ds^2}\left(s^2+a^2\right)^{-1} = a\frac{d}{ds}\left[(-1)\left(s^2-a^2\right)^{-2}2s\right] = -2a\frac{d}{ds}\left[\frac{s}{(s^2+a^2)^2}\right]$$
$$= -2a\left(\frac{\left(s^2+a^2\right)\cdot 1 - s\cdot 2\cdot\left(s^2+a^2\right)\cdot 2s}{(s^2+a^2)^4}\right)$$
$$= -2a\frac{\left(s^2+a^2\right)\left[s^2+a^2-4s^2\right]}{(s^2+a^2)^4} = -2a\frac{\left[-3s^2+a^2\right]}{(s^2+a^2)^3} = 2a\frac{\left[a^2-3s^2\right]}{(s^2+a^2)^3}$$

■

Problem 5.11 Find $L\left[t^2e^{-3t}\sin 2t\right]$

Solution.

$$L\left[t^2e^{-3t}\sin 2t\right] = \frac{d^2}{ds^2}\left[L\left[e^{-3t}\sin 2t\right]\right] = \frac{d^2}{ds^2}\left[\sin 2t\right]_{s\to s+3} = \frac{d^2}{ds^2}\left[\frac{2}{(s+3)^2+4}\right]$$
$$= \frac{d^2}{ds^2}\left[\frac{2}{s^2+6s+13}\right] = 2\frac{d^2}{ds^2}\left(s^2+6s+13\right)^{-1}$$
$$= 2\frac{d^2}{ds^2}\left[\left[(-1)(s^2+6s+13)\right]^{-2}(2s+6)\right]$$
$$= -2\frac{d}{ds}\left[\frac{2s+6}{(s^2+6s+13)^2}\right] = -4\frac{d}{ds}\left[\frac{s+3}{(s^2+6s+13)^2}\right]$$
$$= -4\frac{(s^2+6s+13)^2-(s+3)(s^2+6s)(2s+6)}{(s^2+6s+13)^4}$$
$$= -4\frac{s^2+6s+13-4(s^2+6s+9}{(s^2+6s+13)^3}$$
$$= -4\frac{s^2+6s+14-4s^2-36-24s}{(s^2+6s+13)^3}$$
$$= -4\frac{-3s^2-18s-23}{(s^2+6s+13)^3} = 4\frac{3s^2+18s+23}{(s^2+6s+13)^3}$$

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Theorem 5.2.5 . If $L[f(t)] = F[s]$ and if $\lim_{t\to 0}\frac{f(t)}{t}$ exists, then $L\left[\frac{f(t)}{t}\right] = \int_0^\infty F(s)ds$.

Worked Examples

Problem 5.12 Find $L\left[\frac{\sin 2t}{t}\right]$.

Solution.

$$L\left[\frac{\sin 2t}{t}\right] = \int_s^\infty L[\sin 2t]ds = \int_s^\infty \frac{2}{s^2+4}ds = 2\frac{1}{2}\left(\tan^{-1}\frac{s}{2}\right)_s^\infty = \tan^{-1}\infty - \tan^{-1}\left(\frac{s}{2}\right)$$
$$= \frac{\pi}{2} - \tan -1\left(\frac{s}{2}\right) = \cot^{-1}\left(\frac{s}{2}\right)$$

■

Problem 5.13 Find the Laplace transform of $\frac{e^{at}-e^{-bt}}{t}$.

Solution.

$$L\left[\frac{e^{at}-e^{-bt}}{t}\right] = \int_s^\infty L\left[e^{at}-e^{bt}\right]ds = \int_s^\infty \left(\frac{1}{s-a} - \frac{1}{s+b}\right)ds$$
$$= [\log(s-a) - \log(s+b)]_s^\infty = \left[\log\frac{s-a}{s+b}\right]_s^\infty = \log\frac{s+b}{s-a}$$

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5.2.3 Laplace transform of periodic functions

A function f (t) is said to be periodic if there exists a positive constant T such that $f(t+T) = f(t)$for all t. The smallest of such T is called the period of the function.
Example: $\sin(2t+2\pi) = \sin t$.Sine function is a periodic function with period 2π.

Theorem 5.2.6 If $f(t)$ is a periodic function with period 2π

$$L[f(t)] = \frac{1}{1-e^{-sT}}\int_0^T e^{-st}f(t)dt.$$

Worked Example

Problem 5.14 Find the Laplace transform for the triangular wave function defined by

$$f(t) = \begin{cases} t & \text{for } 0 < t < a \\ 2a - t & \text{for } a < t < 2a \text{ and f is of period a} \end{cases}$$

Solution. Since $f(t)$ is a period $2a$, we have

$$L[f(t)] = \frac{1}{1-e^{-2as}}\int_{2a}^0 e^{-st}f(t)dt$$

$$= \frac{1}{1-e^{-2as}}\left[\int_a^0 e^{-st} f(t)dt + \int_{2a}^{a} e^{-st}(ft)dt\right]$$

$$= \frac{1}{1-e^{-2as}}\left[\int_a^0 e^{-st} tdt + \int_a^{2a} e^{-st}(2a-t)\,dt\right]$$

$$= \frac{1}{1-e^{-2as}}\left[\int_a^0 e^{-st} td\left(\frac{e^{-st}}{-s}\right) + \int_a^{2a} e^{-st}(2a-t)\,d\left(\frac{e^{-st}}{-s}\right)\right]$$

$$= \frac{1}{1-e^{-as}}\left[\left(t\frac{e^{-st}}{-s}\right)_a^0 - \int_a^0 \frac{e^{-st}}{-s}dt + \left((2a-t)\frac{e^{-st}}{-s}\right)_a^{2a} - \int_a^{2a}\frac{e^{-st}}{-s}(-dt)\right]$$

$$= \frac{1}{1-e^{-as}}\left[a\frac{e^{-as}}{-s} + \frac{1}{s}\left(\frac{e^{-st}}{-s}\right)_0^a + 0 + \frac{a}{s}e^{-as} - \frac{1}{s}\left(e^{-st}-s\right)_a^{2a}\right]$$

$$= \frac{1}{1-e^{-as}}\left[-a\frac{e^{-as}}{s} - \frac{1}{s^2}\left(e^{-as-1}\right) + \frac{a}{s}e^{-as} + \frac{1}{s^2}\left(e^{-2as}-e^{-as}\right)\right]$$

$$= \frac{1}{1-e^{-as}}\left[-a\frac{e^{-as}}{s} - \frac{1}{s^2}e^{-as} + \frac{1}{s^2}a\frac{e^{-as}}{s} + \frac{1}{s^2}e^{-2as} - \frac{1}{s^2}\left(e^{-as}\right)\right]$$

$$= \frac{1}{1-e^{-as}}\left[\frac{e^{-2as}-e^{-as}-e-as+1}{s^2}\right] = \frac{1}{1-e^{-as}}\left[\frac{1-2e^{-2as}+e^{-2as}}{s^2}\right]$$

$$= \frac{1}{(1-e^{-as})(1+e^{-as})}\frac{(1-e^{-as})^2}{s^2} = \frac{(1-e^{-as})}{s^2(1+e^{-as})}$$

$$= \frac{e^{\frac{as}{2}}}{-e^{\frac{-as}{2}}}s^2\left(e^{\frac{as}{2}}+e^{\frac{-as}{2}}\right) = \frac{1}{s^2}\tanh\left(\frac{as}{2}\right)$$

■

Problem 5.15 Find the Laplace transform of the half-sine wave rectifier function defined by $f(t) = \begin{cases} \sin\omega t & 0 < t < \frac{\pi}{\omega} \\ 0 & \frac{\pi}{\omega} < t < \frac{2\pi}{\omega} \end{cases}$.

Solution. $f(t)$ is defined in the interval $\left(0, \frac{2\pi}{\omega}\right)$

$$f\left(t+\frac{2\pi}{\omega}\right) = \sin\omega\left(t+\frac{2\pi}{\omega}\right) = \sin(\omega t + 2\pi) = \sin\omega t = f(t)$$

$\therefore$ $f(t)$ is periodic with period $T = \frac{2\pi}{\omega}$.

We know that, $L[f(t)] = \frac{1}{1-e^{-sT}}\int_0^T f(t)e^{-st}dt = \frac{1}{1-e^{-\frac{s2\pi}{\omega}}}\int_0^{\frac{2\pi}{\omega}} e^{-st}\sin\omega t dt$

$$= \frac{1}{1-e^{-\frac{s2\pi}{\omega}}}\left[\frac{e^{-st}}{s^2+w^2}[-s\sin\omega t - \omega\cos\omega t]\right]_0^{\frac{\pi}{\omega}}$$

$$= \frac{1}{1-e^{-\frac{s2\pi}{\omega}}}\left[\frac{e^{\frac{-s\pi}{\omega}}}{s^2+w^2}[-s.0-\omega(-1)] - \frac{-\omega}{\omega^2+s^2}\right]$$

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$$= \frac{1}{\left(1-e^{-\frac{s2\pi}{\omega}}\right)(s^2+\omega^2)}\left\{\omega e^{\frac{-s\pi}{\omega}}+\omega\right\} = \frac{\omega\left(1+e^{-\frac{s\pi}{\omega}}\right)}{(s^2+\omega^2)\left(1-e^{\frac{-s\pi}{\omega}}\right)\left(1+e^{\frac{-s\pi}{\omega}}\right)}$$
$$= \frac{\omega}{(s^2+\omega^2)\left(1-e^{\frac{-s\pi}{\omega}}\right)}.$$

■

Problem 5.16 Find the Laplace transform of the saw-toothed wave function of period T given by $f(t) = \frac{t}{T}, 0 < t < T$.

Solution. Since $f(t)$ is a periodic function of period T, we have

$$L[f(t)] = \frac{1}{1-e^{-sT}}\int_0^T e^{-st}f(t)dt = \frac{1}{1-e^{-sT}}\int_0^T e^{-st}\frac{t}{T}dt = \frac{1}{T(1-e^{-sT})}\int_0^T td\left(\frac{e^{-st}}{-s}\right)$$
$$= \frac{1}{T(1-e^{-sT})}\left[\left(\frac{te^{-st}}{-s}\right)_0^T - \int_0^T \frac{e^{-st}}{-s}dt\right] = \frac{1}{T(1-e^{-sT})}\left[\frac{Te^{-sT}}{-s}+\frac{1}{s}\left(\frac{e^{-st}}{-s}\right)_0^T\right]$$
$$= \frac{1}{sT(1-e^{-sT})}\left[-Te^{-sT}-\frac{1}{s}\left(e^{-sT}-1\right)\right] = \frac{1}{sT(1-e^{-sT})}\left[\frac{1}{s}-\frac{1}{s}e^{-sT}-Te^{-sT}\right]$$
$$= \frac{1}{sT(1-e^{-sT})}\left[\frac{1-e^{-sT}}{s}-Te^{-sT}\right] = \frac{1}{Ts^2}-\frac{e^{-sT}}{s(1-e^{-sT})}$$

■

5.3 Laplace Transforms of Derivatives

Theorem 5.3.1 Let $f(t)$ be a continuous function for all $t \geq 0$ and $f'(t)$ is piecewise continuous on every finite interval $0 \leq t \leq a$ in $[0,\infty)$. If f(t) and $f'(t)$ are of exponential order as $t \to \infty$, then $L[f's(t)] = sL[f(t)] - f(0)$.

Theorem 5.3.2 Find $L[f''(t)]$.

Note 5.3.3 $L[f'''(0)] = s^3L[f(t)] - s^2f(0) - sf'(0) - f''(0)$.
In general $L[f^{(n)}] = s^nL[f(t)] - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - sf^{(n-2)}(0) - f^{(n-1)}(0)$.

Note 5.3.4 The above results will be used later in solving differential equations using Laplace transforms.

5.4 Laplace transform of an integral

If f(t) is piecewise continuous in every finite interval $0 \leq t \leq a$ in $[0,\infty)$ and f(t) is of exponential order $\alpha > 0$ and if L[f(t)] = F(s) then

$$L\left[\int_0^t f(t)dt\right] = \frac{F(s)}{s} = \frac{L[f(t)]}{s} for s > \alpha.$$

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Note 5.4.1 $L\left[\int_0^t \int_0^t f(t)dtdt\right] = \frac{L\left[\int_0^t f(t)dt\right]}{s} = \frac{L[f(t)]}{s^2}$.

Problem 5.17 Find $L[e^{-t}\int_0^t \cos t dt]$.

Solution.

$$\begin{aligned}
L[e^{-t}\int_0^t \cos t dt] &= L\left[\int_0^t t\cos t dt\right]_{s\to s+1} = \left[\frac{L[t\cos t]}{s}\right]_{s\to s+1} \\
&= \frac{1}{s+1}\left(-\frac{d}{ds}\{L(\cos t)\}\right)_{s\to s+1} \\
&= -\frac{1}{s+1}\frac{d}{ds}\left\{\frac{s+1}{s^2+2s+2}\right\} \\
&= -\frac{1}{s+1}\left\{\frac{(s^2+2s+s)-(s+1)(2s+2)}{(s^2+2s+2)^2}\right\} \\
&= -\frac{1}{s+1}\left\{\frac{(s^2+2s+s)-2(s+1)^2}{(s^2+2s+2)^2}\right\} \\
&= -\frac{1}{s+1}\left\{\frac{(s^2+2s+2-2s^2-4s-2)}{(s^2+2s+2)^2}\right\} \\
&= -\frac{1}{s+1}\left\{\frac{-s^2-2s}{(s^2+2s+2)^2}\right\} \\
&= \frac{s(s+2)}{(s+1)(s^2+2s+2)^2}.
\end{aligned}$$

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5.4.1 Inverse Laplace Transform by the method of partial fractions

Worked Examples

Problem 5.18 Find the inverse Laplace transform of $\frac{1}{(s+1)(s+2)}$

Solution.

Let $$\frac{1}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2} = \frac{A(s+2)+B(s+1)}{(s+1)(s+2)}$$

$\therefore$ $$1 = A(s+2)+B(s+1).$$

when $s=-1, A=1$.
when $s=-2, -B=1 \Rightarrow B=-1$.

$\therefore$ $$\frac{1}{(s+1)(s+2)} = \frac{1}{s+1} - \frac{1}{s+2}$$

$$L^{-1}\left[\frac{1}{(s+1)(s+2)}\right] = L^{-1}\left[\frac{1}{s+1} - \frac{1}{s+2}\right] = L^{-1}\left[\frac{1}{s+1}\right] - L^{-1}\left[\frac{1}{s+2}\right] = e^{-t} - e^{-2t}.$$

■

Problem 5.19 Find $L^{-1}\left[\frac{1}{s(s+3)^3}\right]$.

Solution.

Let $$\frac{1}{s(s+3)^3} = \frac{A}{s} + \frac{B}{s+3} + \frac{C}{(s+3)^2} + \frac{D}{(s+3)^3}$$
$$1 = A(s+3)^3 + Bs(s+3)^2 + Cs(s+3) + Ds.$$

When $s = 0, 1 = 27A \Rightarrow A = \frac{1}{27}$.
When $s = -3, 1 = -3D \Rightarrow B = \frac{-1}{3}$.
Equating the coeff. of s^3 we get
$A + B = 0 \Rightarrow B = -A = \frac{-1}{27}$.
Equating the coeff. of s^2 we get

$$9A + 6B + C = 0 \quad \Rightarrow \quad 9 \times \frac{1}{27} + 6\left(\frac{-1}{27}\right) + C = 0 \quad \Rightarrow \quad \frac{9}{27} - \left(\frac{6}{27}\right) + C = 0$$
$$\Rightarrow \quad \frac{3}{27} + C = 0 \quad \Rightarrow \quad C = -\frac{1}{9}$$

$$\therefore \quad \frac{1}{s(s+3)^3} = \frac{\frac{1}{27}}{s} - \frac{\frac{1}{27}}{s+3} - \frac{\frac{1}{9}}{(s+3)^2} + \frac{\frac{1}{3}}{(s+3)^3}$$
$$L^{-1}\left[\frac{1}{s(s+3)^3}\right] = \frac{1}{27}L^{-1}\left[\frac{1}{s}\right] - \frac{1}{27}\left[\frac{1}{s+3}\right] - \frac{1}{9}\left[\frac{1}{(s+3)^2}\right] - \frac{1}{3}\left[\frac{1}{(s+3)^3}\right]$$
$$= \frac{1}{27} \times 1 - \frac{1}{27}e^{-3t} - \frac{1}{9}e^{-3t}t - \frac{1}{3}e^{-3t}\frac{t^2}{2!}$$
$$= \frac{1}{27} - \frac{e^{-3t}}{27} - \frac{e^{-3t}t}{9} - \frac{t^2e^{-3t}}{6}.$$

■

Problem 5.20 Find $L^{-1}\left[\frac{s}{(s-4)(s+4)}\right]$.

Solution. Let $\frac{s}{(s-4)(s+4)} = \frac{A}{s-4} + \frac{B}{s+4}$ $s = A(s+4) + B(s-4)$
Put $s = 4 \Rightarrow A = \frac{1}{2}$
Put $s = -4 \Rightarrow B = \frac{1}{2}$

$$L^{-1}\left[\frac{s}{(s-4)(s+4)}\right] = L^{-1}\left[\frac{\frac{1}{2}}{s-4} + \frac{\frac{1}{2}}{s+4}\right] = \frac{1}{2}e^{4t} + \frac{1}{2}e^{-4t} = \cosh 4t$$

■

Problem 5.21 Find $L^{-1}\left[\frac{5s+3}{(s-1)(s^2+2s+5)}\right]$.

Solution.

Let $$\frac{5s+3}{(s-1)(s^2+2s+5)} = \frac{A}{s-1} + \frac{Bs+C}{s^2+2s+5}$$
$$5s+3 = A\left(s^2+2s+5\right) + (Bs+C)(s-1)$$

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Put $s=1 \Rightarrow 8A=8 \Rightarrow A=1$.
Equating the coeff. $s^2 \Rightarrow A+B=0 \Rightarrow B=-1$.
$s=0 \Rightarrow 5A-C=3 \Rightarrow C=5-3=2$.

$$L^{-1}\left[\frac{5s+3}{(s-1)(s^2+2s+5)}\right]=L^{-1}\left[\frac{1}{s-1}+\frac{-s+2}{s^2+2s+5}\right]=e^t-L^{-1}\left[\frac{s-2}{(s+1)^2+5-1}\right]$$
$$=e^t-L^{-1}\left[\frac{s+1-3}{(s+1)^2+4}\right]=e^t-e^{-t}L^{-1}\left[\frac{s-3}{s^2+4}\right]$$
$$=e^t-e^{-t}\left[\cos 2t-\frac{3}{2}\sin 2t\right]=e^t-\frac{e^{-t}}{2}[2\cos 2t-3\sin 2t].$$

■

5.4.2 Inverse Laplace transform by the method of convolution

Definition 5.4.1 — Convolution. Let $f(t)$ and $g(t)$ be two functions defined for all $t \geq 0$. The convolution of $f(t)$ and $g(t)$ is defined as

$$(f*g)(t)=f(t)*g(t)=\int_0^t f(u)g(t-u)du \text{ or } (f*g)(t)=\int_0^t g(u)f(t-u)du.$$

Theorem 5.4.2 If $L[f(t)]=F(s)$ and $L[g(t)]=G(s)$ then $L[f(t)*g(t)]=F(s)\cdot G(s)$. Also $L^{-1}[F(s)G(s)]=f(t)*g(t)=L^{-1}[F(s)]*L^{-1}[G(s)]$.

Proof. By definition $(f*g)(t)=\int_0^t f(u)g(t-u)du$

$$L[f(t)*g(t)]=\int_0^\infty e^{-st}(f(t)*g(t))dt=\int_0^\infty e^{-st}\left[\int_0^t f(u)g(t-u)du\right]dt$$
$$=\int_0^\infty\int_0^t e^{-st}f(u)g(t-u)dudt. \quad (1)$$

In the above double integration, the region of integration is given by $u=0, u=t$, $t=0$ and $t=\infty$.

We shall evaluate the double integral by changing the order of integration. Divide the region in to strips parallel to the t-axis. PQ is one such strip. Along this strip t varies from u to ∞ and as this strip traverses the entire region, u

varies from 0 to ∞.

$$\therefore \qquad L[f(t)*g(t)]=\int_{u=0}^\infty\int_{t=u}^\infty e^{-st}f(u)g(t-u)dtdu.$$

Let $v=t-u$. When $t=u, v=0$.
$dv=dt$ When $t=\infty, v=\infty$.

$$L[f(t)*g(t)]=\int_{u=0}^\infty\int_{v=0}^\infty e^{-s(u+v)}f(u)g(v)dvdu=\int_{u=0}^\infty\int_{v=0}^\infty e^{-su}e^{-sv}f(u)g(v)dvdu$$

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$$= \int_{u=0}^{\infty} e^{-su} f(u)du \int_{v=0}^{\infty} e^{-sv} g(v)dv = L[f(u)]L[g(v)]$$
$$= L[f(t)]L[g(t)] = F(s)G(s).$$
$$\therefore \quad L^{-1}[f(t)g(t)] = f(t) * g(t).$$

■

Worked Examples

Problem 5.22 Using convolution theorem, evaluate $L^{-1}\left[\frac{1}{(s+1)(s+2)}\right]$.

Solution.

$$\frac{1}{(s+1)(s+2)} = \frac{1}{s+1} \cdot \frac{1}{s+2}$$

Let $F(s) = \frac{1}{s+1}$ and $G(s) = \frac{1}{s+2}$.

$$L^{-1}\left[\frac{1}{(s+1)(s+2)}\right] = L^{-1}[F(s)G(s)] = L^{-1}[F(s)] * L^{-1}[G(s)]$$
$$= L^{-1}\left[\frac{1}{s+1}\right] * L^{-1}\left[\frac{1}{s+2}\right] = e^{-t} * e^{-2t}$$
$$= \int_0^t e^{-u} e^{-2(t-u)} du = \int_0^t e^{-u} e^{-2t+2u} du = \int_0^t e^{-2t+u} du = \int_0^t e^{-2t} e^{u} du$$
$$= e^{-2t} \cdot (e^u)_0^t = e^{-2t}\left(e^t - 1\right) = e^{-t} - e^{-2t}.$$

■

Problem 5.23 Find $L^{-1}\left[\frac{s}{(s^2+a^2)^2}\right]$ using convolution theorem.

Solution.

Let $$\frac{s}{(s^2+a^2)^2} = \frac{s}{s^2+a^2} \cdot \frac{1}{s^2+a^2}$$

$$\therefore \quad F(s) = \frac{s}{s^2+a^2}, G(s) = \frac{1}{s^2+a^2}$$

$$L^{-1}\left[\frac{s}{(s^2+a^2)^2}\right] = L^{-1}[F(s)G(s)] = L^{-1}F(s) * L^{-1}G(s)$$
$$= L^{-1}\left[\frac{s}{(s^2+a^2)}\right] * \left[\frac{1}{(s^2+a^2)}\right] = \cos at * \frac{\sin at}{a}$$
$$= \frac{1}{a}\int_0^t \cos au \sin a(t-u)du = \frac{1}{2a}\int_0^t 2\sin(at-au)\cos au du$$
$$= \frac{1}{2a}\int_0^t \{\sin(at-au+au) + \sin(at-au-au)\}du$$

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$$= \frac{1}{2a}\left\{\int_0^t \sin at\,du + \int_0^t \sin(at-2au)du\right\}$$
$$= \frac{1}{2a}\left[\sin at(u)_0^t - \left[\frac{\cos(at-2au)}{-2a}\right]_0^t\right]$$
$$= \frac{1}{2a}\left[t\sin at + \frac{1}{2a}(\cos(-at)-\cos at)\right]$$
$$= \frac{1}{2a}\left[t\sin at - \frac{1}{2a}0\right] = \frac{t\sin at}{2a}.$$

■

Problem 5.24 Find $L^{-1}\left[\frac{s^2}{(s^2+4)^2}\right]$ using convolution theorem.

Solution.

$$L^{-1}\left[\frac{s^2}{(s^2+4)^2}\right] = L^{-1}\left[\frac{s}{s^2+4}\cdot\frac{s}{s^2+4}\right] = L^{-1}\left[\frac{s}{s^2+4}\right] * L^{-1}\left[\frac{s}{s^2+4}\right]$$
$$= \cos 2t * \cos 2t = \int_0^t \cos 2u\cdot\cos 2(t-u)du$$
$$= \int_0^t \cos 2u\cos(2t-2u)du$$
$$= \int_0^t \left(\frac{\cos(2u+2t-2u)+\cos(2u-2t+2u)}{2}\right)du$$
$$= \frac{1}{2}\int_0^t (\cos 2t + \cos[4u-2t])du = \frac{1}{2}\left[\cos 2t\int_0^t du + \int_0^t \cos(4u-2t)du\right]$$
$$= \frac{1}{2}\left[\cos 2t[u]_0^t + \left[\frac{\sin(4u-2t)}{4}\right]_0^t\right]$$
$$= \frac{1}{2}\left[t\cos 2t + \frac{1}{4}\{\sin 2t - \sin(-2t)\}\right] = \frac{1}{2}\left[t\cos 2t + \frac{1}{4}2\sin 2t\right]$$
$$= \frac{1}{2}\left[t\cos 2t + \frac{\sin 2t}{2}\right] = \frac{1}{4}(\sin 2t + 2t\cos 2t).$$

■

Problem 5.25 Find $L^{-1}\left[\frac{1}{(s+1)(s^2+2s+2)}\right]$ using convolution theorem.

Solution.

$$L^{-1}\left[\frac{1}{(s+1)(s^2+2s+2)}\right] = L^{-1}\left[\frac{1}{(s+1)((s+1)^2+1)}\right] = e^{-t}L^{-1}\left[\frac{1}{s(s^2+1)}\right]$$
$$= e^{-t}\left[L^{-1}\left(\frac{1}{s}\cdot\frac{1}{s^2+1}\right)\right] = e^{-t}\left[L^{-1}\left(\frac{1}{s}\right) * L^{-1}\left(\frac{1}{s^2+1}\right)\right]$$

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$$= e^{-t}[1 * \sin t] = e^{-t}\int_0^t \sin u \cdot 1\,du$$
$$= e^{-t}(-\cos u)_0^t = -e^{-t}(\cos t - 1) = e^{-t}(1 - \cos t).$$

■

Problem 5.26 Using convolution theorem, find $L^{-1}\left[\frac{4}{(s^2+2s+5)^2}\right]$.

Solution.

$$L^{-1}\left[\frac{4}{(s^2++2s+5)^2}\right] = L^{-1}\left[\frac{4}{(s+1)^2+4)^2}\right] = e^{-t}L^{-1}\left[\frac{4}{(s^2+4)^2}\right]$$

$$\text{Now, } L^{-1}\left[\frac{4}{(s^2+4)^2}\right] = L^{-1}\left[\frac{2}{s^2+4}\frac{2}{s^2+4}\right] = L^{-1}\left[\frac{2}{s^2+4}\right] * L^{-1}\left[\frac{2}{s^2+4}\right]$$

$$= \sin 2t * \sin 2t = \int_0^t \sin 2u \sin 2(t-u)\,du$$

$$= \frac{1}{2}\int_0^t (\cos\{2u - 2(t-u)\} - \cos\{2u + 2t - 2u\})\,du$$

$$= \frac{1}{2}\left[\int_0^t \cos(4u - 2t)\,du - \int_0^t \cos 2t\,du\right]$$

$$= \frac{1}{2}\left[\left(\frac{\sin(4u-2t)}{4}\right)_0^t - \cos 2t\,(u)_0^t\right]$$

$$= \frac{1}{2}\left[\frac{\sin 2t}{4} + \frac{\sin 2t}{4} - t\cos 2t\right] = \frac{1}{2}\left[\frac{\sin 2t}{2} - t\cos 2t\right].$$

From(1) $L^{-1}\left[\frac{1}{(s^2++2s+5)^2}\right] = \frac{1}{4}[\sin 2t - 2t\cos 2t]e^{-t}$ ■

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5.5 Solution of linear second order differential equations

Worked Examples

Problem 5.27 Solve $y'' + 5y' + 6y = 2$ given $y'(0) = 0$ and $y(0) = 0$ using Laplace transform method.

Solution. $y'' + 5y' + 6y = 2$.
Taking Laplace transform both sides we get

$$L\left[y''\right] + 5L\left[y'\right] + 6L[y] = L[2]$$
$$s^2L[y] - sy(0) - y'(0) + 5[sL[y] - y(0)] + 6L[y] = 2 \cdot L[1]$$
$$L[y]\left[s^2 + 5s + 6\right] = \frac{2}{s}$$
$$L[y] = \frac{2}{s(s^2+5s+6)} = \frac{2}{s(s+2)(s+3)}$$
$$y = L^{-1}\left[\frac{2}{s(s+2)(s+3)}\right]$$

Let $$\frac{2}{s(s+2)(s+3)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+3}$$
$$= \frac{A(s+2)(s+3) + Bs(s+3) + Cs(s+2)}{s(s+2)(s+3)}$$
$$\therefore \quad 2 = A(s+2)(s+3) + Bs(s+3) + Cs(s+2).$$

When $$s = 0, 6A = 2 \Rightarrow A = \frac{1}{3}.$$
$$\therefore \quad \frac{2}{s(s+2)(s+3)} = \frac{1}{3}\cdot\frac{1}{s} - \frac{1}{s+2} + \frac{2}{3}\cdot\frac{1}{s+3}$$

Now $$y = L^{-1}\left[\frac{2}{s(s+2)(s+3)}\right]$$
$$= L^{-1}\left[\frac{1}{3}\cdot\frac{1}{s} - \frac{1}{s+2} + \frac{2}{3}\cdot\frac{1}{s+3}\right]$$
$$= \frac{1}{3}L^{-1}\left[\frac{1}{s}\right] - L^{-1}\left[\frac{1}{s+2}\right] + \frac{2}{3}L^{-1}\left[\frac{1}{s+3}\right]$$
$$= \frac{1}{3}\times 1 - e^{-2t} + \frac{2}{3}e^{-3t}$$
$$y = \frac{1}{3} - e^{-2t} + \frac{2}{3}e^{-3t}$$

■

Problem 5.28 Solve $\left(D^2+5D+6\right)y = e^{-t}$ given that $y(0) = 0, y'(0) = 0$ using Laplace transforms.

Solution.

Given $$\left(D^2+5D+6\right)y = e^{-t}$$
$$\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = e^{-t}$$

Taking Laplace transform on both sides we get

$$L\left[\frac{d^2y}{dt^2}\right] + 5L\left[\frac{dy}{dt}\right] + 6L[y] = L\left[e^{-t}\right]$$
$$s^2L[y] - sy(0) - y'(0) + 5[sL(y) - y(0)] + 6L[y] = \frac{1}{s+1}$$
$$s^2L[y] + 5sL(y) + 6L[y] = \frac{1}{s+1}$$
$$L[y]\left(s^2+5s+6\right) = \frac{1}{s+1}$$
$$L[y] = \frac{1}{(s+1)\left(s^2+5s+6\right)}$$
$$y = L^{-1}\left[\frac{1}{(s+1)\left(s^2+5s+6\right)}\right]$$
$$= L^{-1}\left[\frac{1}{(s+1)(s+2)(s+3)}\right]$$

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Let $$\frac{1}{(s+1)(s+2)(s+3)} = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{s+3}.$$

$$1 = A(s+2)(s+3) + B(s+1)(s+3) + C(s+1)(s+2)$$

When $s = -1$, $1 = A(1)(2)$ $A = \frac{1}{2}$

When $s = -2$ $1 = B(-1)(1)$ $B = -1.$

When $s = -3$ $1 = C(-2)(-1)$ $C = \frac{1}{2}.$

$\therefore$ $$\frac{1}{(s+1)(s+2)(s+3)} = \frac{\frac{1}{2}}{s+1} - \frac{1}{s+2} + \frac{\frac{1}{2}}{s+3}$$

$$L^{-1}\left[\frac{1}{(s+1)(s+2)(s+3)}\right] = \frac{1}{2}L^{-1}\left[\frac{1}{s+1}\right] - L^{-1}\left[\frac{1}{s+2}\right] + \frac{1}{2}L^{-1}\left[\frac{1}{s+3}\right]$$

$$y = \frac{1}{2}e^{-t} - e^{-2t} + \frac{1}{2}e^{-3t}.$$

■

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